

# **INDUSTRIAL INTERNSHIP REPORT**

## **Quality Prediction in a Mining Process**

**Data Science & Machine Learning**

Industrial Internship Project

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### **Project Objective**

To develop a machine-learning regression approach for predicting % *Silica Concentrate* from process measurements collected from an iron-ore flotation process.

Final internship submission

## Executive Summary

This internship project focuses on **quality prediction in an iron-ore mining flotation process**. The objective is to use process measurements to predict the percentage of silica concentrate, an important quality indicator in the beneficiation process. The project follows a practical data-science workflow: understanding the dataset, cleaning and preprocessing the measurements, preparing the data for modelling, training a supervised regression model, and evaluating its performance.

The dataset used in the completed experiment contains **737,453 records and 24 columns**, including a date field, process variables, iron feed/concentrate measurements, and silica feed/concentrate measurements. For the fast-track implementation, the timestamped numerical measurements were prepared at an hourly level before modelling. An **XGBoost Regressor** was trained and evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and  $R^2$ .

Two experiments were performed. The first used all available numerical process features and achieved **MAE = 0.375573, RMSE = 0.517834, and  $R^2 = 0.793037$** . The second removed % Iron Concentrate to investigate the UCT problem statement concerning the strong relationship between iron and silica concentration. Its performance decreased to **MAE = 0.686353, RMSE = 0.910098, and  $R^2 = 0.360725$** .

The results indicate that the selected process variables can provide useful predictive information for silica concentration and that % Iron Concentrate is an important predictive variable in this implementation. The project also provided practical experience in data preprocessing, exploratory analysis, supervised learning, model evaluation, interpretation of regression metrics, and GitHub-based project submission.

# Table of Contents

|     |                                          |
|-----|------------------------------------------|
| 1   | Preface                                  |
| 2   | Introduction                             |
| 2.1 | About UniConverge Technologies Pvt. Ltd. |
| 2.2 | About upskill Campus                     |
| 2.3 | Objective                                |
| 2.4 | Reference                                |
| 2.5 | Glossary                                 |
| 3   | Problem Statement                        |
| 4   | Existing and Proposed Solution           |
| 5   | Proposed Design / Model                  |
| 5.1 | High-Level Design                        |
| 5.2 | Low-Level Data Flow                      |
| 5.3 | Interfaces / Outputs                     |
| 6   | Performance Test                         |
| 6.1 | Test Plan / Test Cases                   |
| 6.2 | Test Procedure                           |
| 6.3 | Performance Outcome                      |
| 7   | My Learnings                             |
| 8   | Future Work Scope                        |
| 9   | Conclusion                               |
| 10  | References                               |

# 1. Preface

The internship in Data Science & Machine Learning provided an opportunity to connect classroom concepts with a practical machine-learning workflow. During the internship, the learning progression moved from fundamentals of Big Data, Data Science and Machine Learning to data handling, probability and statistics, and finally supervised machine learning and model evaluation.

The final project was selected from the industrial problem statements provided during the program. The chosen problem, Quality Prediction in a Mining Process, was suitable for applying regression concepts to an industrial dataset while remaining aligned with the learning completed during the internship.

The project helped convert theoretical concepts into an end-to-end implementation: loading a real dataset, examining its structure, cleaning numerical values, preparing time-based observations, selecting the target, training a regression model, evaluating predictions, and presenting the findings.

I would like to express my gratitude to the internship mentors, coordinators, and learning resources that supported the completion of this work. The experience strengthened my understanding of how data science projects are structured and how machine-learning results should be interpreted rather than treated as isolated numbers.

## 2. Introduction

Data Science and Machine Learning combine data collection, preprocessing, statistical reasoning, modelling, evaluation and communication to solve practical problems. Industrial processes generate many measurements from sensors, control systems and laboratory observations. Analysing these measurements can support quality monitoring and more informed operational decisions.

The internship's learning sequence covered Big Data and Data Science concepts, the data-science lifecycle, data retrieval and cleaning, exploratory data analysis, probability and statistics, and machine-learning fundamentals. The project applies these concepts to an industrial quality-prediction problem.

### 2.1 About UniConverge Technologies Pvt. Ltd.

UniConverge Technologies Pvt. Ltd. (UCT) is presented in the supplied internship report template as an organisation working across technology-oriented areas including digital transformation, IoT, cybersecurity, cloud technologies, machine learning, communications and software development. The internship context therefore provides exposure to technologies that connect software, data and real-world applications.

### 2.2 About upskill Campus

The supplied internship materials describe upskill Campus and The IoT Academy as the learning and internship ecosystem through which students receive structured technical exposure, resources, project problem statements and practical assignments.

### 2.3 Objective

- Gain practical exposure to a complete Data Science and Machine Learning workflow.
- Apply data preprocessing, exploratory analysis and supervised learning to a real industrial dataset.
- Build a regression model for predicting % Silica Concentrate.
- Evaluate model quality using MAE, RMSE and  $R^2$ .
- Investigate the effect of removing % Iron Concentrate from the feature set.
- Document the implementation and results in a reproducible GitHub-based project.

### 2.4 Reference

The internship learning progression used the weekly study resources and the UCT project problem statement. The final project follows the structure requested in the supplied report template: background, problem statement, existing and proposed solution, design/model, performance testing, learning outcomes and future scope.

### 2.5 Glossary

| Term | Meaning                                                         |
|------|-----------------------------------------------------------------|
| ML   | Machine Learning                                                |
| MAE  | Mean Absolute Error; average absolute prediction error.         |
| RMSE | Root Mean Squared Error; penalises larger errors more strongly. |

|                |                                                                                                   |
|----------------|---------------------------------------------------------------------------------------------------|
| R <sup>2</sup> | Coefficient of determination; indicates the proportion of target variance explained by the model. |
| Feature        | An input variable used by the model.                                                              |
| Target         | The variable the model is trained to predict.                                                     |
| Flotation      | A mineral-processing stage used to separate valuable material from unwanted material.             |
| XGBoost        | A gradient-boosted decision-tree machine-learning algorithm.                                      |

### 3. Problem Statement

The selected UCT problem is **Quality Prediction in a Mining Process**. The project concerns an iron-ore flotation plant and focuses on predicting **% Silica Concentrate** from process measurements.

The problem statement asks whether silica concentration can be predicted at short intervals, how far ahead such prediction can reasonably be made, and whether prediction remains possible without using **% Iron Concentrate**, given the high correlation between the two quality variables.

#### **Problem objectives addressed in this implementation:**

- 1 Build a supervised regression model capable of predicting % Silica Concentrate from available process measurements.
- 2 Evaluate prediction quality using MAE, RMSE and  $R^2$ .
- 3 Compare a model using all available numerical features with a model that excludes % Iron Concentrate.
- 4 Document the limitations of the fast-track implementation and identify improvements for future work.

A methodological limitation should be noted: the completed implementation aggregates timestamped numerical measurements at an hourly level and uses a standard train/test split. Therefore, this report does **not** claim a validated minute-ahead forecasting horizon. Forecasting horizon analysis and time-aware validation are included as future work.

## 4. Existing and Proposed Solution

### 4.1 Existing Approaches

Quality prediction in industrial processes can be approached using statistical models, regression algorithms, tree-based ensembles and, where temporal behaviour is central, dedicated time-series or sequence models. The supplied UCT problem statement motivates a data-driven approach because the process contains many simultaneously measured variables.

### 4.2 Limitations of a Basic Approach

- Using a single random train/test split does not fully represent future production conditions.
- Hourly aggregation reduces the original minute-level temporal resolution.
- A simple feature-based model does not explicitly model lagged process effects.
- The implementation does not yet quantify a validated prediction horizon.
- Model explainability and operational deployment were outside the fast-track scope.

### 4.3 Proposed Solution

The proposed solution is an XGBoost regression pipeline. Numerical process measurements are cleaned and converted into numeric form, timestamps are parsed and sorted, hourly preparation is performed, missing values are handled through median imputation, and an XGBoost Regressor is trained. The target is % Silica Concentrate.

### 4.4 Value Addition

- Transforms a large industrial dataset into a machine-learning-ready representation.
- Provides quantitative prediction-error metrics rather than only descriptive analysis.
- Directly tests the importance of % Iron Concentrate through a controlled feature-removal experiment.
- Produces reusable code and result files suitable for GitHub-based project documentation.

### 4.5 GitHub Submission

Project code: <https://github.com/utk3136/upskillcampus/blob/main/src/main.py>

Final report path reserved in the repository: [GitHub report link](#)



## 5. Proposed Design / Model

### 5.1 High-Level Design

|                                                      |                                                                                |                                                                                |                                                              |                                                                   |
|------------------------------------------------------|--------------------------------------------------------------------------------|--------------------------------------------------------------------------------|--------------------------------------------------------------|-------------------------------------------------------------------|
| <b>Raw Mining Data</b><br>737,453 rows<br>24 columns | <b>Cleaning &amp; Preprocessing</b><br>numeric conversion<br>timestamp parsing | <b>Hourly Data Preparation</b><br>median aggregation<br>missing-target removal | <b>XGBoost Regressor</b><br>80/20 split<br>median imputation | <b>Prediction &amp; Evaluation</b><br>MAE • RMSE • R <sup>2</sup> |
|------------------------------------------------------|--------------------------------------------------------------------------------|--------------------------------------------------------------------------------|--------------------------------------------------------------|-------------------------------------------------------------------|

Figure 1. High-level project pipeline. The arrows represent the logical flow from raw process measurements to the final quality prediction and evaluation.

### 5.2 Low-Level Data Flow

- 1 Load MiningProcess\_Flotation\_Plant\_Database.csv and inspect shape and column names.
- 2 Strip column-name whitespace and convert comma-decimal object values to numeric values where applicable.
- 3 Parse the date column and sort observations chronologically.
- 4 Aggregate numerical process measurements hourly using the median and remove rows without the target.
- 5 Set % Silica Concentrate as the target and use remaining numeric columns as model inputs.
- 6 Split the prepared dataset into training and testing subsets.
- 7 Use median imputation for missing feature values.
- 8 Train the XGBoost Regressor and generate predictions on the test set.
- 9 Calculate MAE, RMSE and R<sup>2</sup>.
- 10 Repeat the experiment after removing % Iron Concentrate.

### 5.3 Interfaces / Outputs

The implementation is command-line based. The main program is stored in **src/main.py**. The project generates a processed dataset, model-results table and visual outputs such as a correlation heatmap, silica distribution and actual-versus-predicted plots. These outputs support the analysis presented in the report.

## 6. Performance Test

### 6.1 Test Plan / Test Cases

| Test Case | Purpose                          | Expected Result                                               |
|-----------|----------------------------------|---------------------------------------------------------------|
| TC-01     | Load and inspect dataset         | 737,453 × 24 dataset loaded successfully                      |
| TC-02     | Prepare numerical/hourly data    | Model-ready data produced                                     |
| TC-03     | Train with all features          | Regression predictions and metrics generated                  |
| TC-04     | Train without % Iron Concentrate | Second set of predictions and metrics generated               |
| TC-05     | Compare experiments              | Effect of removing the correlated quality variable quantified |

### 6.2 Test Procedure

The dataset was successfully loaded with a raw shape of **737,453 rows × 24 columns**. The implementation prepared the numerical measurements at an hourly level and then trained the XGBoost regressor. The completed run reported 820 test observations for each experiment.

MAE was used to quantify the average absolute difference between predicted and actual silica concentration. RMSE was used to give greater weight to larger prediction errors.  $R^2$  was used to assess how much variation in the target was explained by the model relative to a baseline based on the target mean.

### 6.3 Performance Outcome

| Experiment                 | Model             | MAE      | RMSE     | $R^2$    | Test rows |
|----------------------------|-------------------|----------|----------|----------|-----------|
| With all features          | XGBoost Regressor | 0.375573 | 0.517834 | 0.793037 | 820       |
| Without % Iron Concentrate | XGBoost Regressor | 0.686353 | 0.910098 | 0.360725 | 820       |

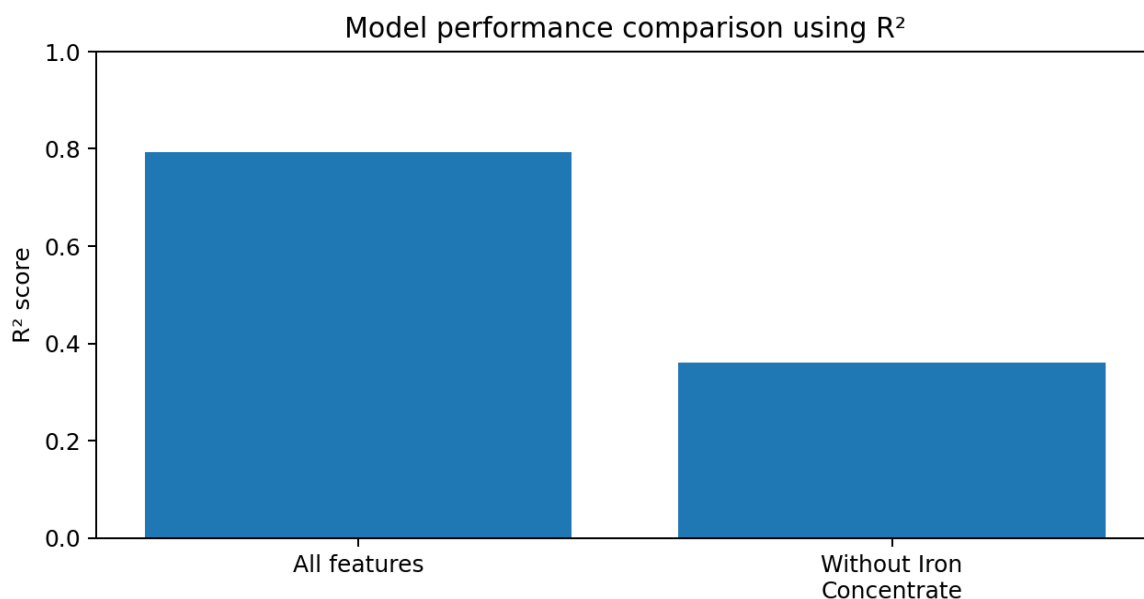


Figure 2.  $R^2$  comparison based on the actual completed model run.

**Interpretation:** The all-feature experiment achieved an  $R^2$  of approximately 0.793, indicating that the model explained a substantial portion of target variation in the held-out test data used by this

implementation. After removing % Iron Concentrate,  $R^2$  decreased to approximately 0.361. MAE increased by about 82.8% and RMSE increased by about 75.7%, showing a substantial loss in predictive performance.

## 7. My Learnings

The project connected the concepts studied throughout the internship with a practical machine-learning task.

### Technical Learnings

- Understanding a large real-world dataset by inspecting rows, columns, data types and target variables.
- Handling numerical values represented using decimal commas and converting them into machine-readable numeric values.
- Working with timestamps and preparing time-indexed industrial measurements.
- Understanding the difference between raw data volume and the smaller modelling dataset produced after aggregation.
- Building a supervised regression pipeline with preprocessing and an XGBoost model.
- Using MAE, RMSE and  $R^2$  together rather than relying on a single evaluation metric.
- Understanding how removing a highly informative feature can affect predictive performance.

### Analytical Learnings

- A strong model score must be interpreted in the context of the data preparation and validation strategy.
- Feature importance and correlation can influence model design, but correlation alone does not establish causation.
- Industrial prediction problems often require temporal validation because future observations differ from randomly selected observations.
- Clear documentation of assumptions and limitations is an important part of a data-science project.

### Professional Learnings

- Organising a project into code, requirements, documentation and generated outputs.
- Using Git and GitHub for version-controlled project submission.
- Communicating model results in a concise, evidence-based manner.
- Planning future improvements instead of overstating what the current implementation proves.

## 8. Future Work Scope

The current implementation is intentionally focused on a working baseline. Several extensions could make the solution more suitable for a production-oriented industrial forecasting system.

- **Time-aware validation:** replace the random split with chronological train/validation/test splits or rolling-window validation.
- **Prediction horizon analysis:** explicitly construct lag and lead features to investigate minute-ahead and hour-ahead prediction.
- **Lag features:** include previous observations of key process variables to represent process dynamics.
- **Feature engineering:** derive rolling means, standard deviations, rates of change and process ratios where scientifically justified.
- **Model comparison:** compare XGBoost with Random Forest, Gradient Boosting, linear regression and other suitable baselines.
- **Hyperparameter tuning:** optimise tree depth, learning rate, estimators and sampling parameters using validation data.
- **Explainability:** use feature-importance and SHAP-style analysis to understand which process variables influence predictions.
- **Deployment:** package the model in a simple dashboard/API so that process data can be entered and predictions viewed interactively.
- **Monitoring:** track model drift and prediction error after deployment.

These extensions would also allow the original UCT questions about prediction frequency and forecasting horizon to be investigated more rigorously.

## 9. Conclusion

This internship project successfully implemented a machine-learning workflow for quality prediction in an iron-ore mining flotation process. The completed run processed a dataset containing 737,453 records and 24 columns and trained an XGBoost regression model to predict % Silica Concentrate.

The all-feature experiment produced  $MAE = 0.375573$ ,  $RMSE = 0.517834$  and  $R^2 = 0.793037$  on the reported test set. Removing % Iron Concentrate reduced performance to  $MAE = 0.686353$ ,  $RMSE = 0.910098$  and  $R^2 = 0.360725$ . Within the scope of this implementation, the comparison demonstrates that % Iron Concentrate contributes important predictive information.

The project should be viewed as a practical baseline rather than a production-ready forecasting system. The next stage would be time-aware validation, explicit lag/horizon modelling, broader model comparison and deployment. Overall, the work demonstrates the ability to take an industrial problem statement through data preparation, machine-learning implementation, evaluation and technical documentation.

## 10. References

- 1 UniConverge Technologies Pvt. Ltd. / upskill Campus internship project problem statement: Quality Prediction in a Mining Process.
- 2 Davy Cielen, Arno D. B. Meysman and Mohamed Ali, *Introducing Data Science: Big Data, Machine Learning, and More, Using Python Tools*.
- 3 Vijay K. Rohatgi and A. K. Md. Ehsanes Saleh, *An Introduction to Probability and Statistics*.
- 4 Alex Smola and S. V. N. Vishwanathan, *Introduction to Machine Learning*.
- 5 Kaggle dataset: *Quality Prediction in a Mining Process*, used as the source of the mining flotation process dataset.
- 6 Project GitHub repository: <https://github.com/utk3136/upskillcampus>
- 7 Project source code: <https://github.com/utk3136/upskillcampus/blob/main/src/main.py>
- 8 Final report repository path: [https://github.com/utk3136/upskillcampus/blob/main/QualityPredictionMiningProcess\\_Utkarsh\\_USC\\_UCT.pdf](https://github.com/utk3136/upskillcampus/blob/main/QualityPredictionMiningProcess_Utkarsh_USC_UCT.pdf)

**Submission note:** The performance values reported in this document are the values produced by the completed project run supplied during the internship workflow. No benchmark metrics from other repositories have been used as project results.